What you will learn in this chapter

What you will learn in this chapter:

- That **objective complexity** cannot be computed, only approximated.
- That **probability** can be defined algorithmically.
- That **randomness** can only be defined algorithmically.
- That algorithmic information sets limits to **the power of AI** (Gödel's theorem).

Detailed outline Chapter (3)

Algorithmic information applied to mathematics

Kolmogorov complexity is not computable! But...

Kolmogorov complexity appears as an ideal notion that can be proven to be incomputable. However,

- it is a fundamental mathematical notion,
- and it can be approximated for practical (and realistic) use.

Algorithmic probability and algorithmic information

Probability can be defined in purely algorithmic terms.
This, however, requires a distinction between two versions of Kolmogorov complexity.

Randomness

What is randomness? The only sound definition of randomness is based on Kolmogorov complexity.

Deductive reasoning

Complexity arguments can be used to prove theorems.

Gödel's theorem revisited

Gödel's theorem is one of the most important theorems in mathematics. It sets limits to what any formal theory can prove (and indirectly, to what AI can achieve). Thanks to Kolmogorov complexity, Gödel's theorem becomes almost trivial, and much easier to grasp.
